

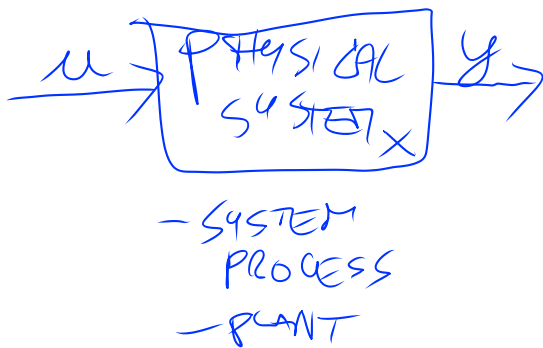
<https://sites.google.com/diag.uniroma1.it/liberati/home/teaching/2026-systems-and-control-methods-for-cyber-physical-security>

For open questions: also insert formulas, not just verbal descriptions. For exercises, write formulas as well as explanations.

A project is possible. Up to three papers needs to be studied, we do a report and simulations of the paper, and then we present the project at the oral exam, which becomes mandatory in this specific case, and theory questions will be asked as well.

What are we going to see in the course?

What is a cyber physical system? It's a physical system, which can be also called System Process, Plant.



We're going to describe/model a system in mathematical terms.

It will have:

- input variables u_a , things that we can use to influence the physical system.
- output variables y : variables which we want to control (like the velocity of a car, the position).

Note that u and y are vectors of input elements. Thus, we use u to influence y .

- state variables x , set of all the variables that perfectly characterise the current state of a system. By knowing x , we know everything about our system. x are like internal variables. The values can be set at any time.

Typically the time variable is not specified, but it's just for simplicity.

$$u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{n_u} \end{pmatrix}, x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n_x} \end{pmatrix}, y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n_y} \end{pmatrix}$$

In a cyber physical system, we may have thousands of variables, with the input we act only on a limited number of variables.

The part in which we give inputs is known as the actuation part, while the output part is the observation part.

We want to measure the output as well, and we can do this with sensors. From the sensor block, connected to $y(t)$, we output $y_m(t)$, the measured value, also known as the measured output.

Typically the output variables are a subset of the state variables. Not all of them, as measuring every variable could be problematic. So, we can see only a small part of the actual system.

After we've measured a variable, we send $y_m(t)$ to an object, the Controller. It could be a human or an algorithm. The controller takes, as inputs, two things: $y_m(t)$ as said before, as well as the references, $y_{\text{ref}}(t)$. In an algorithm designing a car, the references could be the maximum velocity, and so on. We want to achieve control, which means that the output variables of the system are as close as possible to the desired reference. The output of the controller is the action, what we need to do with inputs. The command $u(t)$ will be sent to the system, to the Actuator module, which turns commands as the physical input $u_a(t)$.

Alongside all of this, we also have two other types of inputs to the system: w and n . They're exogenous inputs we cannot control. w is for disturbances, while n are noise variables. They capture two different things. Disturbance is everything that has high physical impact on the system (like wind on the car), Noise is related to measurements noise n_y , as sensors are not perfect. Noise is problematic, as y_m used by the controller will be partially "wrong" (like the speed of the car is 50Km/h, while with noise it could be 51Km/h). Noises are then low energy, but high impact.

We've modelled a close loop, measuring what we want to control, and acting accordingly. The left part is known as input channel, while the right part is measurement channel.

One fundamental problem we need to solve in the Controller is state estimation, to properly decide which controls we should apply for our system. The problem is that the state vector is huge, but what we can measure is small. So state estimation is the problem of finding $\hat{x}(t)$, the state estimate. If we have a good estimate of the state of the system, we can build powerful controllers. State estimation in physical centres is probably the most critical part.

The first attack we're going to see in detail is an attack on state estimation.

But before, let's complete the picture of the system. We need once another object, the Monitor, which is a system located close to the controller, and its goal is to observe anything, reading the value of the sensors (y_m), and will take a look at the decisions of the controller $u(t)$. The monitor typically has some knowledge of the system, given by a model of the system. The output of the monitor is given by two options:

- attack/fault detection (boolean, yes or no)
- identification, determining which part of the system is under attack.

The down part is the cyber part/algorithms, and it's digital. While the upper part is analog. The links between the two parts are known as communication channels.

So, in a cyber physical system we always have:

- digital object
- communication network
- physical part

This takes the name of networked cyber physical system.

Where an attacker could hack? The first investigations dealt with what could happen with attacks taking place directly on the physical system. A fault is something that might just happen. The controller that we design are robust with respect to faults.

An Hacker will want to launch powerful attacks, and needs to be studied.

Attackers could also do an attack on the communication channels, corrupting data, summing an exogenous value a_y on the measured output, as well as an attack modifying a command the controller is sending to the physical system, a_u .

So, the value of y_m could also be called y_m^a , which states that the measured values are corrupted. A corrupted measurement will lead to corrupt state estimation, and the controller will take bad decisions.

So, to recap the attack possibilities:

- direct physical attack on system
- false data injection attack on input/output channels.
- Modification of controller/detector parameters.

A "dumb" attack is quite easily detected by a monitor.

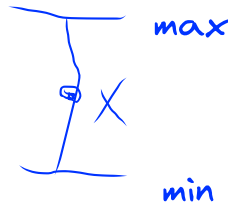
The attacker will have two main goals:

1. Cause damage
2. Remain undetected, the monitor should keep on saying zero, even with no attacks.

The defender of the system is the operator of the system. Then we have the attacker as the other party.

How to model attack at high level?

1. Damage: the focus is on state variables (variables which completely characterise the state of the system). Typically, in the real world we want to keep safe operativity of the system, so all state variables must remain within acceptable bounds. So, for every state variable x_i , we (as operators) must respect (keep) the variable within safety upper and lower limits (x_i^{min}, x_i^{max}).



This is a box constraint.

So, for the entire state vector we can identify a region, S_x , the safety region, saying that the state can evolve, but must always remain within S_x .

The goal of the attacker is to bring the state outside the safety limit.

The attacker could carefully then inject false data to lead the state of the system outside of the safety limits. What happens in practice when $x(t) \notin S_x$?

Two different things:

- 1: Physical Damage.
- 2: Tripping of protections: close to the physical objects we have local monitors monitoring physical values, and to prevent physical damage they might disconnect the object when detecting something strange. This is a win for the attacker as well, as we have loss of service and cascading failures, potentially causing economical damage.

2) Remain undetected as long as possible. Let's call the output of the monitor as $\phi_1(t)$, which could be true or false. ϕ_1 is the output of the monitor, and not necessarily the reality (false positives or false negatives). The monitor works

using the residual based detection as typical technique. Internally, the monitor computes a residual signal $r(t)$, such that, typically:

$$\begin{aligned} r(t) &\approx 0 \text{ in normal/nominal operation} \\ r(t) &\text{ large in case the monitor sees there is something wrong} \end{aligned}$$

How can we frame undetectability? The attacker must carefully select f_a (physical attack), a_u , a_y in order to have a maximum possible input on the state $x(t)$, while minimizing the impact on the residual. So, the residual $r(t)$ might remain low, and still have an attack, an undetected attack.

Is it possible, then, to have undetectable attacks?